



GaitAI Proposal / Report

Surveillance Intelligence | Gait Biometrics | Medical Motion Analytics

MobilityCare Sprint 1

WalkScan + FallRisk Research-to-MVP Blueprint

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GaitAI MobilityCare Sprint 1

WalkScan + FallRisk Research-to-MVP Blueprint

This report is the first structured assignment for Pankhuri as Research Scientist in the GaitAI healthcare sector. It gives one complete end-to-end task and breaks it into a simple 1-week sprint with outputs, rules, and acceptance criteria.

Sprint goal

By the end of the week, Pankhuri should deliver a practical blueprint for a video-based healthcare gait-analysis MVP that can convert a short walking video into mobility metrics, fall-risk screening logic, and a clinical-style report template.

Executive Summary

The first assignment should focus on GaitAI MobilityCare, specifically WalkScan and FallRisk. WalkScan will convert a short walking video into measurable gait and mobility indicators. FallRisk will use gait, balance, variability, posture, and longitudinal trends to categorize risk as low, medium, or high. The first week is not for model training or medical claims. It is for research grounding, feature definition, dataset mapping, MVP workflow, scoring logic, and report design.

The work must be easy for engineering, product, and clinical stakeholders to use. The final output should become the starting document for the first GaitAI healthcare prototype.

Key decisions for Week 1

- Primary product focus: WalkScan and FallRisk only.
- Primary input: 10-20 second walking video, preferably side-view and full body visible.
- Primary output: mobility score, fall-risk category, key gait metrics, and PDF report template.
- Validation stance: screening and research support only, not diagnosis.
- Engineering readiness: output should specify features, pipeline, datasets, and prototype plan.

1. First End-to-End Task

Task title

GaitAI MobilityCare: WalkScan + FallRisk Research-to-MVP Blueprint

Objective

Prepare a complete research and product blueprint for a healthcare gait-analysis system that takes a short walking video and generates a clinical-style mobility report. The output should support two GaitAI healthcare products: WalkScan, a camera-based gait assessment report, and FallRisk, a fall-risk screening and prevention intelligence module.

Scope

- Understand the healthcare problem and target users: physiotherapists, hospitals, elderly-care centers, neurologists, caregivers, home-care teams, and rehab clinics.
- Identify gait features that can be extracted from video and/or wearables.
- Map public datasets and important papers useful for healthcare gait analysis.
- Design a first MVP workflow from video input to report output.
- Design an explainable v1 fall-risk scoring logic.
- Create a sample clinical report template for GaitAI WalkScan.
- Prepare final recommendations and a 30-day roadmap.

Expected final folder

She should submit one folder named GaitAI_MobilityCare_Sprint_1_Pankhuri with all documents, tables, diagrams, and the final 5-slide summary.

File	Purpose
01_GaitAI_Healthcare_Usecase.md	Problem statement, target users, product need, and healthcare use case notes.
02_Clinical_Gait_Features.xlsx	Feature table with meaning, importance, video-extractability, and use case.
03_Dataset_and_Literature_Review.xlsx	Public datasets and 10 important papers mapped for GaitAI relevance.
04_WalkScan_MVP_Pipeline.pdf	Simple architecture diagram from video to PDF report/dashboard.
05_FallRisk_Scoring_Logic.md	Rule-based v1 scoring logic with limitations and validation requirements.
06_WalkScan_Report_Template.pdf	One-page clinical-style sample report for doctors/physiotherapists.
07_Final_Recommendations.md	What GaitAI should build next and open questions.
README.md	Folder index, summary, and how to review the sprint outputs.

2. One-Week Sprint Plan

The sprint is divided into daily outputs. Each day should end with one visible deliverable so progress is easy to review.

Day	Focus	Clear Work	End-of-Day Output
Day 1	GaitAI healthcare use case	Study GaitAI MobilityCare, WalkScan, FallRisk, RehabTrack, WatchCare, elderly care, physiotherapy, hospitals, neurology, and home care.	1-2 page note: What GaitAI MobilityCare Should Solve in Healthcare.
Day 2	Clinical gait parameters	Research cadence, step/stride time, walking speed, symmetry, trunk sway, posture, arm swing, balance, variability, and fall-risk signs.	Clinical gait feature table with meaning, importance, extractability from video, and use case.
Day 3	Datasets and literature	Find public datasets and 10 important papers on healthcare gait analysis, fall-risk, Parkinsonian gait, stroke, rehab, IMU, and video-based pose.	Dataset table and literature summary table.
Day 4	WalkScan MVP design	Design video upload, pose estimation, skeleton tracking, gait feature extraction, risk scoring, and report generation workflow.	MVP pipeline diagram and input/output specification.
Day 5	FallRisk scoring logic	Create explainable v1 scoring using gait speed, cadence, symmetry, variability, trunk sway, posture, and instability indicators.	FallRisk v1 scoring logic document with risk categories and limitations.
Day 6	Clinical report template	Design a one-page WalkScan report that a physiotherapist or clinician can understand quickly.	Sample GaitAI WalkScan Report PDF.
Day 7	Final compilation	Compile all outputs, clean language, prepare summary and next steps.	Final folder + 5-slide summary.

Daily communication rule

At the end of each day, Pankhuri should send: (1) what was completed, (2) what is unclear, (3) files produced, and (4) what she will do next day. Keep updates short and evidence-based.

3. Detailed Instructions: Days 1-3

Day 1 - Understand GaitAI and healthcare problem

- Read the GaitAI website and understand MobilityCare as a healthcare, rehab, sports, elderly-care, and wearable intelligence vertical.
- Write who the users are: physiotherapists, doctors, rehab clinics, elderly-care centers, caregivers, hospitals, neurology clinics, and home-care teams.
- Write the current problem: most gait assessment is subjective, manual, not scalable, and hard to compare across sessions.
- Write the GaitAI answer: objective movement intelligence from video and wearable signals.

Day 2 - Clinical gait parameters

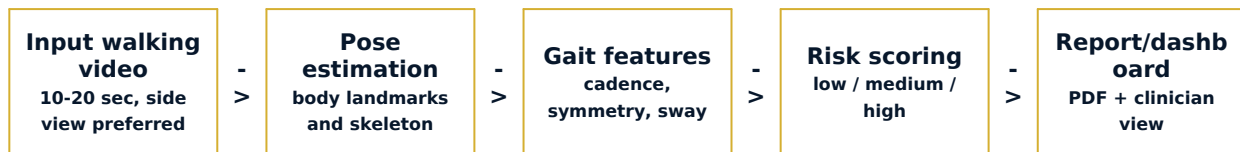
- Make a table of gait features and explain them in simple clinical language.
- Separate video-extractable features from features needing IMU, force plate, or clinical sensors.
- Mark features useful for elderly care, neuro movement, rehab tracking, sports recovery, or general mobility screening.
- Do not copy paper text. Summarize in your own words.

Day 3 - Dataset and literature mapping

- Find public datasets with gait videos, skeletons, wearable signals, inertial sensors, pressure/force plates, or clinical labels.
- Prioritize healthcare-relevant datasets: elderly gait, fall-risk, Parkinsonian gait, post-stroke gait, rehabilitation, orthopedic gait, and daily mobility.
- For every dataset, mention data type, number of subjects, labels, access link, limitations, and whether it is useful for GaitAI.
- Prepare a short table of 10 important papers with method, dataset, key finding, and relevance.

4. Detailed Instructions: Days 4-7

Day 4 - WalkScan MVP design



- Define input: 10-20 second video, side-view preferred, one person, full body visible, good lighting.
- Define processing: pose estimation, skeleton tracking, keypoint smoothing, step detection, gait feature extraction, quality check.
- Define output: skeleton overlay, gait metrics, fall-risk category, mobility score, and downloadable PDF report.
- Mention possible tools: Python, OpenCV, MediaPipe/YOLO pose, NumPy, Pandas, Matplotlib, FastAPI or Streamlit.

Day 5 - FallRisk v1 scoring

- Start with an explainable rule-based score, not a medical diagnostic model.
- Use measurable indicators such as slow walking, reduced cadence, high variability, asymmetry, trunk sway, posture instability, and trend decline.
- Define low, medium, and high risk in plain language.
- Clearly write: This is an AI-assisted mobility screening score and must be clinically validated before medical use.

Day 6 - WalkScan report template

- Create a one-page report with patient/session details, video quality status, mobility score, fall-risk level, key metrics, asymmetry, posture/balance observations, trend placeholder, and disclaimer.
- Use language that a physiotherapist, doctor, caregiver, and patient can understand.
- Do not write diagnosis. Use screening, observation, indicator, recommendation for professional review.

Day 7 - Final compilation

- Clean all files and put them in the final folder.
- Prepare a 5-slide summary: problem, GaitAI solution, features/datasets, MVP pipeline, next 30-day roadmap.
- Make sure every claim is sourced or clearly marked as an assumption.
- Write final recommendations for the engineering team.

5. Feature List for WalkScan and FallRisk

The feature table should tell GaitAI what to extract, why it matters, and whether it can be extracted from a simple walking video.

Feature	Meaning	Why important	Video?	Useful for
Cadence	Steps per minute.	Shows walking rhythm and mobility decline.	Yes	Elderly, rehab, general mobility
Walking speed	Estimated distance/time or pixel-normalized movement speed.	Slow speed is a key mobility and fall-risk signal.	Partly	Elderly, hospital, rehab
Step time	Time between consecutive steps.	Irregular step timing may show instability.	Yes	FallRisk, neuro
Stride time	Time for a full gait cycle.	Useful for rhythm and variability analysis.	Yes	Rehab, neuro
Left-right asymmetry	Difference between left and right movement timing or range.	May indicate injury, stroke, pain, or compensation.	Yes	Rehab, ortho, neuro
Trunk sway	Side-to-side or forward-back upper body motion.	High sway can indicate balance problems.	Partly	FallRisk, elderly
Posture angle	Forward lean or abnormal body alignment.	Postural change can affect stability and fatigue.	Yes	Rehab, Parkinsonian gait
Knee/hip range	Approximate joint motion during gait cycle.	Reduced range can indicate stiffness or weakness.	Partly	Rehab, ortho
Gait variability	Frame-to-frame or step-to-step inconsistency.	High variability often links to instability.	Yes	FallRisk, elderly, neuro
Mobility score	Composite 0-100 summary.	Useful for simple tracking across sessions.	Derived	Clinics, caregivers

Important feature rule

In Week 1, Pankhuri should define features and extraction feasibility. She should not promise clinical accuracy or fixed medical thresholds without validation.

6. Dataset and Literature Review Output

Pankhuri should create a dataset map that tells the team which data can support the first prototype and which data is only useful for background research.

Dataset review fields

Column	What to fill
Dataset name	Official dataset name and source.
Data type	Video, skeleton, wearable/IMU, force plate, pressure mat, clinical record, or multimodal.
Subjects	Number of participants if available.
Condition/setting	Healthy, elderly, Parkinson's, post-stroke, orthopedic, rehab, fall-risk, sports, etc.
Labels	Disease label, fall label, clinical score, step events, gait phase, or no labels.
Access	Open download, request-only, paid, restricted, or not available.
GaitAI usefulness	High / Medium / Low and why.
Limitations	Small dataset, lab-only, sensor-specific, non-video, license restrictions, no clinical labels, etc.

Literature review fields

Paper field	What to capture
Title and year	Full title, publication year, and venue if available.
Problem	What healthcare gait problem the paper solves.
Method	Pose estimation, CNN/LSTM/Transformer, hand-crafted features, IMU model, clinical scoring, etc.
Dataset	Dataset used or data collection setup.
Key result	Summarized result without exaggeration.
Relevance to GaitAI	How this helps WalkScan, FallRisk, RehabTrack, WatchCare, or clinical validation.

7. FallRisk v1 Scoring Logic

The first FallRisk score should be transparent and explainable. It should not be presented as a diagnosis. It should be a screening layer that highlights when a clinician or caregiver should review the person more carefully.

Indicator	Low Risk	Medium Risk	High Risk	Why it matters
Walking speed	Normal for age/context	Slightly slow	Very slow or visibly hesitant	Slow speed can reflect mobility decline and reduced confidence.
Cadence	Stable rhythm	Reduced rhythm	Very low or irregular rhythm	Rhythm reduction can indicate fatigue, weakness, or neuro-motor change.
Left-right symmetry	Balanced	Mild asymmetry	Clear asymmetry	Asymmetry can indicate pain, weakness, injury, stroke, or compensation.
Trunk sway	Stable trunk	Moderate sway	High sway or instability	Greater sway can be related to balance issues.
Gait variability	Low variability	Moderate variability	High variability	Irregular gait can indicate instability and higher fall concern.
Posture	Upright/stable	Mild lean	Strong lean or unstable posture	Poor posture may affect balance and movement efficiency.

Suggested first formula

Use a simple weighted score out of 100. Example: speed 20%, cadence 15%, symmetry 20%, trunk sway 15%, variability 20%, posture 10%. Convert the final score into Low / Medium / High risk bands only after internal testing and clinical review.

Mandatory wording

Disclaimer

FallRisk v1 is an AI-assisted mobility screening score. It is not a medical diagnosis and must be validated with clinicians before real-world medical use.

8. WalkScan Clinical Report Template

Pankhuri should design a one-page WalkScan sample report. The layout should be simple, clinician-friendly, and patient-safe.

Report section	What it should contain
Patient/session details	Patient code, age group if available, date, session number, camera/view, walking distance/time.
Video quality check	Full body visible, lighting, camera view, person count, occlusion, tracking confidence.
Mobility score	0-100 summary score with short interpretation.
Fall-risk level	Low / Medium / High with screening-only wording.
Key gait metrics	Cadence, step time, stride time, walking speed estimate, asymmetry, variability.
Posture and balance	Trunk sway, forward lean, visible instability, joint range notes.
Left-right asymmetry	Difference in left/right timing, range, or stance pattern.
Trend section	Placeholder for comparison with previous sessions.
Recommendation	Professional review, repeat test, rehab follow-up, or caregiver monitoring.
Disclaimer	AI-assisted screening report, not medical diagnosis.

Example report language

The patient shows mild left-right gait asymmetry and moderate trunk sway. A repeat assessment after 2-4 weeks of therapy is recommended to track progress. This report is AI-assisted and should be interpreted by a qualified healthcare professional.

9. Prototype Plan for Engineering

The report should end with a technical plan that can be handed to the engineering team. The prototype should be small but demonstrable.

Prototype v1

Module	What to build first	Output
Upload/capture	Upload a 10-20 second walking video.	Video file accepted and quality checked.
Pose estimation	Run pose model and extract body landmarks frame by frame.	Skeleton keypoints and confidence scores.
Tracking and smoothing	Smooth noisy keypoints and track one walking subject.	Cleaned keypoint time series.
Feature extraction	Calculate step count, cadence, symmetry, trunk sway, posture angle, variability.	Structured feature table.
Scoring	Apply explainable FallRisk v1 logic.	Mobility score and low/medium/high risk.
Visualization	Generate skeleton overlay and simple charts.	Annotated video/image and charts.
Report	Create downloadable WalkScan PDF.	Clinical-style report.
Dashboard	Optional Streamlit/FastAPI dashboard.	Internal demo UI.

Suggested tools

- Python for backend logic.
- OpenCV for video reading, frame handling, and overlays.
- MediaPipe Pose or YOLO pose for first skeleton extraction.
- NumPy and Pandas for time-series calculations.
- Matplotlib for simple charts.
- ReportLab or HTML-to-PDF for PDF report generation.
- Streamlit or FastAPI for internal demo.

10. Working Rules, Quality Bar, and Acceptance Criteria

What she should not do

- Do not start model training before dataset and feature understanding.
- Do not make medical claims or write diagnosis.
- Do not use any patient data without consent and documented approval.
- Do not copy text from papers or websites.
- Do not create fake accuracy numbers or unsupported performance claims.
- Do not overcomplicate the MVP with too many diseases or use cases in Week 1.
- Do not focus on SecureVision/security in this sprint; focus only on healthcare MobilityCare.

Acceptance criteria

Her work is acceptable if GaitAI leadership can clearly answer the following questions after reading her final folder:

- What healthcare problem is GaitAI solving first?
- Which gait features will WalkScan extract?
- Which datasets can support the first prototype?
- How will FallRisk v1 be calculated and explained?
- What will the first clinical report look like?
- What should the engineering team build next?
- What still needs clinical validation, data approval, and product decision-making?

Best final deliverable name

GaitAI MobilityCare Sprint 1: WalkScan and FallRisk Research-to-MVP Blueprint

11. Suggested 30-Day Roadmap After Sprint 1

After Pankhuri completes Sprint 1, the next month can move from research blueprint to a working internal prototype.

Week	Goal	Output
Week 2	Prototype setup	Working video upload, pose extraction, skeleton overlay, and initial feature table.
Week 3	Metric calculation	Cadence, step timing, asymmetry, trunk sway, posture, variability, quality check.
Week 4	Report generation	WalkScan PDF report with mobility score and FallRisk v1 category.
Week 5	Pilot readiness	Internal demo, sample videos, clinician feedback form, validation checklist, consent draft.

Open questions for leadership

- Which clinical partner or physiotherapist will review the report language and feature relevance?
- Will the first MVP be video-only, wearable-only, or video + wearable later?
- What is the minimum demo dataset for internal testing?
- What privacy, consent, and retention rules will GaitAI follow for healthcare videos?
- Which customer segment should be approached first: physiotherapy clinics, elderly-care centers, hospitals, or home-care teams?

End of report